

★ Estimation Theory ➤

Few Definitions ➤

① Estimator:- A rule or method of estimating a population parameter is known as estimator. It is generally expressed as a function of sample variables.

② Estimate- A particular value of an estimator obtained from ~~the~~ set of values of a random sample is known as estimate.

§ (a) Point estimate - based on the sample data, a single number is calculated to estimate the population parameter, which is known as point estimation.

(b) Interval estimator based on the sample data, two sample numbers are calculated to form an interval within which the parameter is expected to lie, this interval is known as interval estimation.

Point Estimation:-

Characteristics or properties of a good point estimate ➤

1. Unbiasedness
2. Consistency
3. Efficiency
4. Sufficiency

1. Unbiasedness - An estimator $\hat{\theta}$ is said to be unbiased estimator, if its expected value is equal to population parameter mean (E)

$$E(\hat{\theta}) = \theta$$

↑ unknown parameters

Baisedness ^{when} $E(\hat{\theta}) \neq \theta$, then it is said baised.

a. a $\Rightarrow$ +vely baised estimator $\Rightarrow E(\hat{\theta}) > \theta$

b. b $\Rightarrow$ -vely baised estimator $\Rightarrow E(\hat{\theta}) < \theta$

Example: - Unbaised estimator: -

$\bar{X}$ = sampling distribution of sample mean

$$\bar{X} = \hat{\theta} \text{ (let)}$$

Expected value of $\bar{X}$ is $E(\bar{X}) = \mu_{\bar{X}} = \mu = \theta$ _{unknown parameters}

$$\Rightarrow E(\hat{\theta}) = \theta$$

So $\{\bar{X} \text{ is an unbaised estimator of population mean}\}$

2. Consistency - An estimator said to be consistent, if its variances approaches to zero, ^{as} ~~and~~ sample size (n) approaches to ∞

variance $\rightarrow 0$ as $n \rightarrow \infty$

$$\text{ie } \text{Var}(\hat{\theta}) \rightarrow 0 \text{ as } n \rightarrow \infty$$

then $\hat{\theta}$ is said Consistent

Example: - $\bar{X} = \hat{\theta}$

$$\text{Var}(\bar{X}) = \sigma_{\bar{X}}^2 = \frac{\sigma^2}{n}$$

$$\text{here } \lim_{n \rightarrow \infty} \frac{\sigma^2}{n} = 0$$

$$\text{ie } \left(\lim_{n \rightarrow \infty} \text{Var}(\bar{X}) = 0 \right) \leftarrow \text{proved}$$

hence $\bar{X}$ is consistent

3. Efficiency: Efficiency is a relative term. Efficiency of an estimator defined by comparing it with another estimator.

if $\hat{\theta}_1, \hat{\theta}_2$ are two unbiased estimator then $\hat{\theta}_1$ is said to be efficient estimator of

① if

$$\text{Var}(\hat{\theta}_1) < \text{Var}(\hat{\theta}_2)$$

* जिसकी भी Variance कम हो, वो efficient है other नहीं.

Q X_1, X_2 and X_3 is a random variable sample of size 3. from a population with mean μ and variance σ^2 . T_1, T_2 and T_3 are estimators used to estimate mean value μ .

Where $T_1 = X_1 + X_2 - X_3$

$$T_2 = 2X_1 + 3X_3 - 4X_2$$

$$T_3 = \frac{1}{3}(\lambda X_1 + X_2 + X_3)$$

Very Imp

→ show $E(T) = \mu$

(i) are T_1 and T_2 unbiased estimator

(ii) find the value of λ such that T_3 is unbiased estimator

(iii) With this value of λ , is T_3 consistent estimator

(iv) Which is the best estimator?

Sol → ① To show $E(T_1) = \mu$ & $E(T_2) = \mu$

$$* E(T_1) = E(X_1 + X_2 - X_3)$$

$$= E(X_1) + E(X_2) - E(X_3)$$

Population ki mean given μ , do we will let sample ki mean bhi $= \mu$

So $E(T_1) = \mu + \mu - \mu$

$$E(T_1) = \mu$$

$$* E(T_2) = E(2X_1 + 3X_3 - 4X_2)$$

$$= 2E(X_1) + 3E(X_3) - 4E(X_2)$$

$$= 2\mu + 3\mu - 4\mu$$

$$E(T_2) = \mu$$

$$\therefore E(T_2) = E(T_1) = \mu \quad \text{ie } T_1 \& T_2 \text{ are unbiased}$$

$$(2) \quad E(T_3) = \mu \quad \text{To prove } T_3 \text{ is unbiased}$$

$$\Rightarrow \frac{1}{3} [\lambda E(X_1) + E(X_2) + E(X_3)] = \mu$$

$$\Rightarrow \frac{1}{3} [\lambda \mu + \mu + \mu] = \mu$$

$$\Rightarrow \lambda \mu + 2\mu = 3\mu$$

$$\Rightarrow \lambda \mu = \mu$$

$$\Rightarrow \lambda = 1$$

So for $\lambda = 1$, T_3 is unbiased

T is unbiased if $E(T) = \mu$

$$(3) \quad \lambda = ? \text{ for } T_3 \text{ consistent estimator}$$

ie $\text{Var} \rightarrow 0$ if sample size $\rightarrow \infty$

$$T_3 = \frac{1}{3} (X_1 + X_2 + X_3) \quad \text{avg}$$

$$= \bar{X}$$

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

$$n \rightarrow \infty, \lim_{n \rightarrow \infty} \frac{\sigma^2}{n} = 0$$

ie $\text{Var}(\bar{X}) \rightarrow 0$ ie T_3 is consistent

(iv) which is Best estimator

$$\begin{aligned}\text{Var}(T_1) &= \text{var}(x_1 + x_2 + x_3) \\ &= \text{var}(x_1) + \text{var}(x_2) + \text{var}(x_3) \\ &= 3\sigma^2\end{aligned}$$

$$\begin{aligned}\text{Var}(T_2) &= \text{var}(2x_1 + 3x_2 - 4x_3) \\ &= 4\text{var}(x_1) + 9\text{var}(x_2) + 16\text{var}(x_3) \\ &= 29\sigma^2\end{aligned}$$

$$\begin{aligned}\text{Var}(T_3) &= \text{var}\left(\frac{1}{3}x_1 + \frac{1}{3}x_2 + \frac{1}{3}x_3\right) \\ &= \frac{1}{9}(\text{var}(x_1) + \text{var}(x_2) + \text{var}(x_3)) \\ &= \frac{3}{9}\sigma^2 \\ &= \frac{1}{3}\sigma^2\end{aligned}$$

Lowest Var is of T_3 , i.e. T_3 is Best estimator

4. Sufficient estimator an estimator is said to be sufficient for a parameter if it contains all the information in the sample regarding the parameter.

Fisher-Neymann Criteria:-

A statistic $T = t(x_1, x_2, \dots, x_n)$ is a sufficient estimator of a parameter if and only if the likelihood function can be expressed as

$$L = \prod_{i=1}^n f(x_i, \theta) = g(T, \theta) h(x)$$

Where,

$g(T, \theta)$ is pdf of statistics T
and $h(x)$ is a function of x only

2-step Approach to find sufficient estimator

1. Define the likelihood function,

$$L = \prod_{i=1}^n f(x_i, \theta) \text{ and simplify it}$$

2. Check whether it can be written in the form of $g(T, \theta) h(x)$

(we have to extract T from L)

Q1 Let $x_1, x_2, \dots, x_n$ be a random sample from a population with pdf of $f(x, \theta) = \theta x^{\theta-1}$ where x ranges from 0 to 1, $\theta > 0$.
find sufficient estimator of θ .

sol $f(x, \theta) = \theta x^{\theta-1}; 0 < x < 1; \theta > 0$

$$L = \prod_{i=1}^n f(x_i, \theta)$$

$$= \prod_{i=1}^n \theta x_i^{\theta-1}$$

$$= \theta^n \prod_{i=1}^n \frac{x_i^{\theta}}{x_i}$$

$$= \underbrace{\theta^n \left(\prod_{i=1}^n x_i \right)^{\theta}}_{g(T, \theta)} \underbrace{\frac{1}{\left(\prod_{i=1}^n x_i \right)}}_{h(x)} \Rightarrow h(x)$$

The function $g(\tau, \theta) = \theta^n \left(\prod_{i=1}^n x_i \right)^\theta$

$$h(x) = \frac{1}{\left(\prod_{i=1}^n x_i \right)}$$

$$T = \prod_{i=1}^n x_i \quad \checkmark$$

Q.2 $f(x, \theta) = \frac{1}{\theta} e^{-x/\theta}, \quad 0 < x, \quad \theta < \infty$

Sol $L = \prod_{i=1}^n f(x_i, \theta)$

$$= \prod_{i=1}^n \frac{1}{\theta} e^{-x_i/\theta}$$

$$= \frac{1}{\theta^n} \left(\prod_{i=1}^n \frac{1}{e^{x_i/\theta}} \right)$$

$$= \frac{1}{\theta^n} \cdot e^{-\frac{\sum x_i}{\theta}}$$

from here $h(x) = 1$

$$g(\tau, \theta) = \frac{1}{\theta^n} \cdot e^{-\frac{\sum x_i}{\theta}}$$

$$T = \sum_{i=1}^n x_i \quad \checkmark$$

Q-3 If $x_1, x_2, \dots, x_n$ are independent and follow poisson distribution with parameter λ , find the sufficient estimator for λ .

Sol

$$f(x, \lambda) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$L = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}$$

$$= e^{-n\lambda} \prod_{i=1}^n \frac{\lambda^{x_i}}{x_i!}$$

$$= \underbrace{e^{-n\lambda} \cdot \lambda^{\sum x_i}}_{g(x)} \cdot \underbrace{\frac{1}{\prod_{i=1}^n x_i!}}_{h(x)}$$

$$g(T, \lambda) = e^{-n\lambda} \cdot \lambda^{\sum x_i}$$

$$\boxed{T = \sum_{i=1}^n x_i} \text{ is the sufficient estimator}$$

Q-4 Let $x_1, x_2, \dots, x_n$ be a random sample from $N(\mu, \sigma^2)$. find the sufficient estimator for μ and σ^2 .

likelihood of μ and σ^2 is
 all coefficient of μ and σ^2 are same for μ and σ^2

★ Method of moment 1

Moment :-

let X be a random variable and $C \in \mathbb{R}$ be a scalar, then the k^{th} moment of X is $E(X^k)$ and k^{th} moment of X about C is $E[(X-C)^k]$

$$E(X^k) = \sum_x x^k p(x) \quad (\text{or}) \quad \int_{-\infty}^{\infty}$$

$$E[(X-C)^k] = \sum_x (x-C)^k p(x) \quad (\text{or}) \quad \int_{-\infty}^{\infty} (x-C)^k f(x) dx$$

$$\mu = E(X) \quad \leftarrow k=1$$

$$C = \mu, \quad k=2$$

$$E[(X-\mu)^2] = \text{Var}(X)$$

Sample Moment :- Let X be a random variable and $C \in \mathbb{R}$ be a scalar

Let $x_1, x_2, \dots, x_n$ be a sample from X

Then the k^{th} moment of X is $\frac{1}{n} \sum_{i=1}^n x_i^k$

$P_k(x_i)$ only as every element has same probability $\frac{1}{n}$ because random sampling

and k^{th} moment about C is $\frac{1}{n} \sum_{i=1}^n (x_i - C)^k$

Method of moments (MOM)

Let $x_1, x_2, \dots, x_n$ be sample observation from PMF $P_X(\theta)$ if X is discrete and PDF $f_X(\theta)$ if X is continuous, where θ

then we define (MOM) estimator

$$\hat{\theta}_{\text{MOM}} \text{ of } \theta = (\theta_1, \theta_2, \dots, \theta_k)$$

to be a solution of k simultaneous equation where for $j=1, 2, \dots, k$ we set the j th true moment to be equal to the j th sample moment

$$E(X) = \frac{1}{n} \sum_{i=1}^n x_i$$

$$E(X^k) = \frac{1}{n} \sum_{i=1}^n x_i^k$$

Q1 Let $x_1, x_2, \dots, x_n$ be the sample observations from X which follows the Exp. distribution with the parameter θ (can be represented as $X \sim \text{Exp}(\theta)$). What is the MOM estimator of θ ?

Sol $f(\theta) = \theta e^{-\theta x}$

$$E(X) = \frac{1}{n} \sum_{i=1}^n x_i$$

$$E(X) = \frac{1}{\theta} = \frac{1}{n} \sum x_i$$

$$\hat{\theta}_{\text{MOM}} = \frac{n}{\sum_{i=1}^n x_i}$$

Q2 of Poisson distribution

$$f(x) = \frac{e^{-\theta} \theta^x}{x!}$$

$$E(x) = \theta = \frac{1}{n} \sum x_i$$

$$\hat{\theta}_{\text{mom}} = \frac{1}{n} \sum_{i=1}^n x_i \quad \checkmark$$

Q3 Uniform distribution from 0 to θ

$$f(x) = \frac{1}{\theta}$$

$$\frac{\theta}{2} = \frac{1}{n} \sum x_i \Rightarrow \frac{2 \sum_{i=1}^n x_i}{n} = \hat{\theta}_{\text{mom}} \quad \checkmark$$

Q4 Normal distribution (θ_1, θ_2) what MOM estimate for θ_1 and θ_2

Q4

$$\theta_1 = \mu$$

$$\theta_2 = \sigma^2$$

$$E(x) = \frac{1}{n} \sum x_i$$

$$\hat{\theta}_{1\text{mom}} = \frac{1}{n} \sum x_i \quad \checkmark$$

$$\text{Var}(x) = E(x^2) - (E(x))^2$$

$$\theta_2 = E(x^2) - \theta_1^2$$

$$\theta_2 + \theta_1^2 = E(x^2)$$

$$\theta_2 + \theta_1^2 = \frac{1}{n} \sum (x_i)^2$$

$$\theta_{2\text{mom}} = \frac{1}{n} \sum (x_i)^2 - \left(\frac{1}{n} \sum x_i \right)^2 \quad \checkmark$$

Bayesian Estimation :-

Case 1: When x is discrete

a. Prior probability: The probability of seeing a particular parameter $P(\theta)$ is known as the ~~prior~~ prior probability.

b. likelihood probability: the likelihood of seeing an outcome given a certain parameter which $P(X/\theta) =$

c. Posterior Probability: The multiplication of prior and likelihood probability
 $P(\theta) \cdot P(X/\theta)$

The Bayesian estimation methods state that the higher the ~~prior~~ posterior ~~prob~~ probability, the better the chance of having that parameter.

Q.1

There are 3 coins one with heads on both sides, one with tails on both sides and a fair coin.

A coin is picked at random and after 3 tosses 2 heads were recorded, find the unknown parameters of getting a head using bayesian estimation.

fair coin = {head, tail}

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 ① p = probability of getting a head

$$P(p=1) = \frac{1}{3}$$

$$P(p=0) = \frac{1}{3}$$

$$P(p=\frac{1}{2}) = \frac{1}{3}$$

} prior probability

$$n=3, x=2$$

$$② \quad P(x=2 | p=1) = p \times p \times (1-p)$$

$$= p^2(1-p)$$

$$\text{(since } p=1)$$

$$= 1^2(1-1)$$

$$= 0$$

$$P(x=2 | p=0) = p^2(1-p)$$

$$= 0$$

$$P(x=2 | p=\frac{1}{2}) = p^2(1-p)$$

$$= \frac{1}{2} \times \frac{1}{4} \Rightarrow \boxed{\frac{1}{8}}$$

③	p	$P(\theta)$	$P(x \theta)$	$P(\theta) \cdot P(x \theta)$
	1	$\frac{1}{3}$	0	0
	0	$\frac{1}{3}$	0	0
	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{8}$	$\frac{1}{24}$

∴ The bayesian estimator of $p \Rightarrow \boxed{\hat{p} = \frac{1}{2}}$ ✓

Q A machine learning Algorithm for computer vision is trained to observe the first vehicle that passes ~~at~~ from an intersection at or after 8am everyday ~~at~~. Then it reports the time from that vehicle to next again and again. We assume this time is Exponentially distributed with unknown parameter λ .

Case-2: Continuous Case

- $\leftarrow f(\theta)$
- a. Prior distribution :- It is the distribution of the continuous parameter θ
- b. Likelihood function :- $L(\theta)$
- c. Posterior Distribution :- multiplication of prior & likelihood function
 $\Rightarrow f(\theta)L(\theta)$

The Bayesian estimation for continuous parameters states that the Bayesian estimation is found by maximizing posterior distribution

Q-1 The mortality risk of a hospital is called the probability of death occurring to any patient admitted to the hospital. It is exponentially distributed with mean 1.5. It has been observed that 25 patients died out of first 50 admitted to hospital. What is Bayesian estimator for the true mortality rate of the hospital.

if death rate is denoted by p

pdf of exp. $f(x) = \lambda e^{-\lambda x}$

mean $= \frac{1}{\lambda} = 1.5$

$\Rightarrow f(p) = \frac{1}{1.5} e^{-\frac{1}{1.5} p}$

$\lambda = \frac{1}{1.5}$

Likelihood func

$L(p) = p^{25} (1-p)^{25}$
25 red 25 blue

Posterior

$f(p) \cdot L(p) = \left(\frac{1}{1.5} e^{-\frac{1}{1.5} p} \right) p^{25} (1-p)^{25} \quad \text{--- (1)}$

maximizing Eq (1) for Bayesian estimates

$$\begin{aligned} \frac{d}{dp} (f(p) \cdot L(p)) &= \left(-\frac{1}{(1.5)^2} e^{-\frac{1}{1.5} p} \right) p^{25} (1-p)^{25} \\ &\quad + \left(\frac{1}{1.5} e^{-\frac{1}{1.5} p} \right) 25 p^{24} (1-p)^{25} \\ &\quad + \left(\frac{1}{1.5} e^{-\frac{1}{1.5} p} \right) p^{25} \cdot 25 (1-p)^{24} (-1) \\ &= \left[\frac{1}{1.5} e^{-\frac{1}{1.5} p} \right] \left\{ -\frac{1}{1.5} p^{25} (1-p)^{25} + 25 p^{24} (1-p)^{25} \right. \\ &\quad \left. - 25 p^{25} (1-p)^{24} \right\} \\ &= \left(\frac{1}{1.5} e^{-\frac{1}{1.5} p} \right) (p^{24} (1-p)^{24}) \left\{ -\frac{1}{1.5} p (1-p) + 25(1-p) \right. \\ &\quad \left. - 25p \right\} \\ &= \left(\frac{1}{1.5} e^{-\frac{1}{1.5} p} \right) (p^{24} (1-p)^{24}) \{ - \end{aligned}$$

$\hat{p} = 0.1666$

Q Assume that we have a population distribution with PDF
 $f(x) = (\theta + 1)x^\theta$ for $0 \leq x \leq 1$.

Assume that θ is not totally random but it is instead
distributed with PDF $f(\theta) = \frac{1}{12}(3 - \theta)$ $-2 \leq \theta \leq 2$.
prior distribution

We have collected a sample of $X_1 = 0.9$, $X_2 = 0.89$,
 $X_3 = 0.76$, $X_4 = 0.96$

What is the Bayesian estimator of θ .

Ans
 $\hat{\theta} = 1.85$

(1) prior distribution

$$f(\theta) = \frac{1}{12}(3 - \theta) \quad ; \quad -2 \leq \theta \leq 2$$

(2) Likelihood function

$$\begin{aligned} L(\theta) &= f(x_1) \times f(x_2) \times f(x_3) \times f(x_4) \\ &= (\theta + 1)(0.9)^\theta \times (\theta + 1)(0.89)^\theta \times \\ &\quad (\theta + 1)(0.76)^\theta \times (\theta + 1)(0.96)^\theta \\ &= \cancel{(\theta + 1)^4} \\ &= \end{aligned}$$

Estimation of Unknown parameter $\rightarrow$

① Point estimation

- $\hookrightarrow$ Maximum Likelihood Method
- $\hookrightarrow$ Method of Moment
- $\hookrightarrow$ Bayesian estimate

② Interval estimate

- $\hookrightarrow$ estimate the mean of parameter

* Interval estimation :-

~~Confidence Interval~~

Confidence Interval (C.I.) $\Rightarrow$ Let X be a random sample, then the confidence Interval for parameter θ with confidence level γ (gamma), is an interval $\rightarrow$ $(L(X), U(X))$ with probability

the property that the probability $\Rightarrow$ $P(L(X) < \theta < U(X)) = \gamma = 1 - \alpha$
 $\uparrow$ Confidence level $\uparrow$ level of Significance

(mostly from 90%, 95% or 99% confidence level γ interval $(L(X), U(X))$)

(eg, if confidence level is $\gamma = 0.95$ (ie 95% confidence level) ie parameter θ $\frac{2}{3}$ of the particular interval of lie data θ $\frac{2}{3}$ of 95% chances $\frac{2}{3}$.)

We need to find

$$P(L(x) < \theta < U(x)) = \gamma$$

(α = level of significance)

~~We consider it to be normally distributed~~

$$P(-Z_{\alpha/2} < Z < Z_{\alpha/2}) = \gamma = 1 - \alpha$$

Interval estimation for mean

$$P\left(-Z_{\alpha/2} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < Z_{\alpha/2}\right) = \gamma = 1 - \alpha$$

let for 95% confidence level i.e. $\gamma = 0.95$, $\alpha = 0.05$

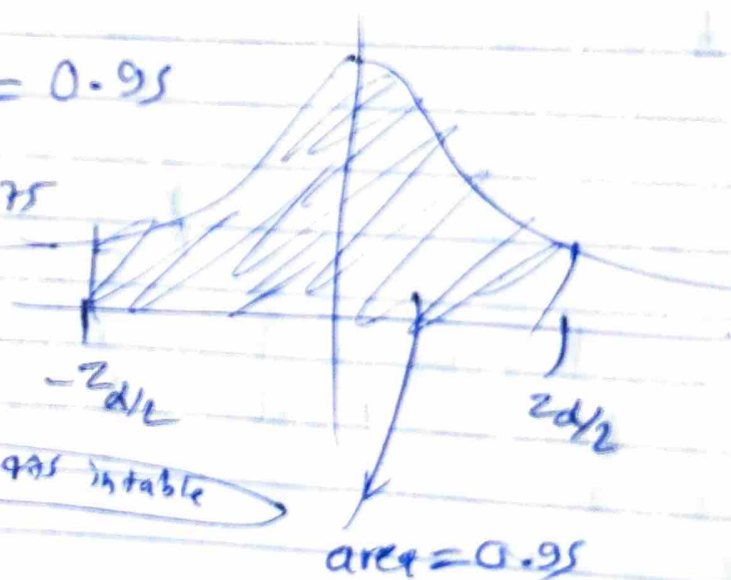
$$P\left(-Z_{\alpha/2} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < Z_{\alpha/2}\right) = 0.95$$

$$2P(0 < Z < Z_{\alpha/2}) = 0.95$$

$$P(0 < Z < Z_{\alpha/2}) = 0.475$$

$$Z_{\alpha/2} = 1.96$$

for $z = 1.96$, we get 0.475 in table



∴ interval

$$\left(\bar{X} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right) \text{ Ans}$$

Q descriptive statistics on a variable ^{measured} ~~measured~~ in a sample of $n=3539$ participants, attending the examination of heart surgery are given

as $n=3539$, $\bar{x} = 127.3$, $\sigma = 19.0$
_{sample mean}

Generate a 95% confidence interval for the unknown mean of the population

Interval $\rightarrow$

$$\left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

$$\left(\frac{127.3 - 1.96 \cdot (19.0)}{\sqrt{3539}}, \frac{127.3 + 1.96 \cdot (19.0)}{\sqrt{3539}} \right)$$

$$(126.673, 127.926)$$

$$(127.7, 127.9)$$

A Random sample of 30 Apple were taken from a large population. On measuring their diameters the mean diameters of sample is $\bar{x} = 91$ mm with $\sigma = 8$ mm, Calculate 95% confidence level for the mean of whole population of sample

$$P(-z_{\alpha/2} < z < z_{\alpha/2}) = 0.95$$

$$P(0 < z < z_{\alpha/2}) = \frac{0.95}{2}$$

$$P(0 < z < z_{\alpha/2}) = 0.475$$

$$z_{\alpha/2} = 1.96$$

the interval

$$\left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

$$\Rightarrow \left(91 - \frac{(1.64)(8)}{\sqrt{30}}, 91 + \frac{(1.64)(8)}{\sqrt{30}} \right)$$

$$\Rightarrow \underline{(88.9, 93.1)} \text{ Ans}$$

If population standard deviation (σ) is unknown,
But sample's standard deviation (s) is known,
then confidence interval $\Rightarrow$

$$\left(\bar{x} - t_{\alpha/2} \frac{s}{\sqrt{n}}, \bar{x} + t_{\alpha/2} \frac{s}{\sqrt{n}} \right)$$

same

Confidence interval μ , σ is unknown:-

The CI for population mean (μ), when population standard deviation is unknown, is given by the interval,

$$\left(\bar{x} - t_{\alpha/2} \frac{s}{\sqrt{n}}, \bar{x} + t_{\alpha/2} \frac{s}{\sqrt{n}} \right)$$

with α level of significance -
& $(n-1)$ ~~level~~ degrees of freedom

★ Hypothesis testing :-

population parameter = population mean = μ unknown = ?

definition of hypothesis: hypothesis is a conjecture about the parameters of population.

If a statement

$$\mu = 5000$$

then it is null hypothesis

& $\mu \neq 5000$
is alternate

Two types of hypothesis

(i) Null hypothesis: a null hypothesis which is actually tested for acceptance or rejection is known as null hypothesis

denoted by H_0

It contains a statement of equality (" $=$ ", "<", ">")

(ii) Alternative hypothesis: a statement of population parameter which gives an alternative to null hypothesis

(i.e.) If H_0 is ^{to be accepted} ~~acceptance~~, which hypothesis has to be rejected and vice versa.

denoted by H_1 or H_A

It contains a statement of inequality (" $\neq$ ", "<", ">")

There are two types of error: - after applying a test, a decision is taken about the acceptance or rejection of null hypothesis. There is always some possibility of committing error.

There are two types of errors:

Type I error:- the error of rejecting the null hypothesis, H_0 .

$\alpha = P(\text{Rejecting } H_0, \text{ even when } H_0 \text{ is true})$

Type II error:- the error of accepting the null hypothesis, H_0 when alternative hypothesis (H_1) is true.

Levels of Significance α is type I error It is the quantity of risk of type I error which we are ready to tolerate in making a decision about null hypothesis (H_0).

denoted by α

Critical Region: The values of test-statistics on which the null hypothesis is rejected is known as the "critical region" or "rejection region".

hypothesis test:- the process that uses sample statistics to test a claim about the values of the population parameter is known as hypothesis test.

Steps in hypothesis testing:-

- (i) Identify H_0, H_1
- (ii) to choose α , (ie level of significance) (given here)
- (iii) to calculate the test statistics
- (iv) determine the Critical Region
- (v) Make a decision.

if a hypothesis comes in critical/rejection region, i.e. claim H_0 is false

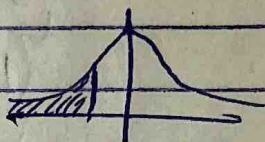
critical region

There are three types of hypothesis test:

- ① Left tailed test
- ② Right tailed test
- ③ two tailed test

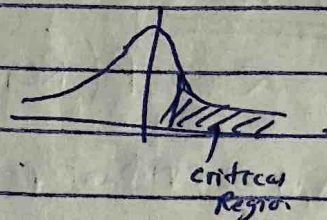
① Left tailed test - if alternative hypothesis (H_1) contains the less than ($<$) inequality, the hypothesis is known as left tailed test of hypothesis

$$H_0: \mu \geq k$$
$$H_1: \mu < k$$



② Right tailed test - if alternative hypothesis (H_1) contains greater than ($>$) inequality, the hypothesis is known as Right tailed test / hypothesis

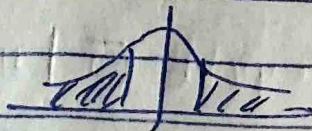
$$H_0: \mu \leq k$$
$$H_1: \mu > k$$



③ Two tailed test when the alternative hypothesis (H_1) contains not equal to sign ($\neq$), the test is known as two tailed test

$$H_0: \mu = k$$

$$H_1: \mu \neq k$$



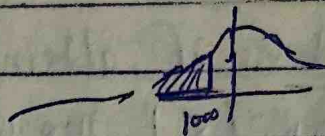
Ex- a manufacturer claims that its rechargeable battery has avg life of atleast 1000 charges

find null & alternate hypothesis -

& type of test

$$H_0: \mu \geq 1000$$

$$H_1: \mu < 1000$$



sign

Left tailed test

Ans

22/Nov/22

Type I error

$$\alpha = P(\text{Rejecting } H_0, \text{ even when } H_0 \text{ is true})$$

if $W \Rightarrow$ critical region

$$\alpha = P(X \in W | H_0)$$

accepting H_0

if H_0 is not

in critical region

Type II error

$$\beta = P(\text{accepting } H_0, \text{ even when } H_1 \text{ is true})$$

$$\beta = P(X \in \bar{W} | H_1)$$

$\bar{W}$ complement

Q8 $P(x \in \bar{W} | H_1) + P(x \in W | H_1) = 1$

$\Rightarrow P(x \in W | H_1) = 1 - P(x \in \bar{W} | H_1)$

$= 1 - \beta$

Power of test

$P(\text{rejecting } H_0, \text{ when } H_1 \text{ is true}) = 1 - \beta$

Q9 given the freq. function $f(x, \theta) = \frac{1}{\theta}$ where $0 \leq x \leq \theta$

and we are testing for the null hypothesis H_0

which $H_0: \theta = 1$

$H_1: \theta = 2$

by means of a single observed value x .

What would be size of type 1 & type 2 error if you choose the interval

(i) $0.5 \leq x$

(ii) $1 \leq x \leq 1.5$

As the critical region

(i) $0.5 \leq x$
 $\alpha = P(x \in W | H_0)$

$\alpha = P(0.5 \leq x | \theta = 1)$

x is lower limit $= 0.5$

as given $0 \leq x \leq \theta$

i.e. upper limit

if $x = \theta = 1$

$\alpha = P(0.5 \leq x \leq 1)$

$$\alpha = \int_{0.5}^1 \frac{1}{\theta} d\theta \Big|_{\theta=1}$$

$$\alpha = \frac{1}{1} \int_{0.5}^1 d\theta$$

$$\alpha = 0.5$$

$$\beta = P\{x \in W | H_1\}$$

$$= P\{x \leq 0.5 | \theta = 2\}$$

$$= \lim_{x \rightarrow 0} P\{x \leq 0.5 | \theta = 2\}$$

$$= P(0 \leq x \leq 0.5)$$

$$= \int_0^{0.5} \frac{1}{\theta} d\theta \Big|_{\theta=2}$$

$$= \frac{1}{2} \int_0^{0.5} d\theta$$

$$= \frac{1}{2} [\theta]_0^{0.5}$$

$$\beta = \frac{0.5}{2} = 0.25$$

(ii) If $1 \leq x \leq 1.5$ under H_0

$$\alpha = P(x \in W | H_0)$$

$$= P(1 \leq x \leq 1.5 | \theta = 1)$$

$$= P(1 \leq x \leq 1)$$

$$= \int_1^{1.5} \frac{1}{\theta} d\theta \Big|_{\theta=1} = \frac{1}{1} [\theta]_1^{1.5} = 0$$

$$\beta = P(\theta = 2 | x \in W | H_1)$$

$$= P(\theta = 2 | x \in W)$$

$$= P(\theta = 2 | x \in W)$$

$$P(\theta = 2 | H_1) = P(\theta = 2 | H_1)$$

$$= 1 - P\{x \in W | H_1\}$$

$$= 1 - P(1 \leq x \leq 1.5 | \theta = 2)$$

$$= 1 - P(1 \leq x \leq 1.5)$$

$$= 1 - \int_1^{1.5} \frac{1}{\theta} d\theta \Big|_{\theta=2}$$

$$= 1 - \frac{1}{2} \int_1^{1.5} d\theta$$

$$= 1 - \frac{1}{2} [1.5 - 1]$$

$$= 1 - \frac{0.5}{2}$$

$$\boxed{\beta = 0.75} \quad \checkmark$$

If $x \geq 1$ is the critical Region for testing null hypothesis $H_0: \theta = 2$, $H_1: \theta = 1$ on the basis of single observation population

$$f(x, \theta) = \theta e^{-\theta x} \quad 0 \leq x < \infty$$

obtain type I & type II error.

$$W: x \geq 1$$

$$\bar{W}: x < 1$$

$$\alpha = P(X \in W | H_0)$$

$$= P(X \geq 1 | \theta = 2)$$

$$= \int (1 \leq x \leq 2)$$

$$= \int_1^{\infty} \theta e^{-\theta x} dx \Big|_{\theta=2}$$

$$= \int_1^{\infty} 2 e^{-2x} dx$$

$$= \left[\frac{2 e^{-2x}}{-2} \right]_1^{\infty} \rightarrow \left[-e^{-2x} \right]_1^{\infty} = -e^{-\infty}$$

$$= \left[-e^{-\infty} + e^{-2} \right]$$

$$= 0 + e^{-2} = \frac{1}{e^2}$$

$$= \underline{\underline{0.1353}}$$

$$\beta = P(X \in W | H_1)$$

$$= P(X < 1 | \theta = 1)$$

$$= P(0 \leq x < 1)$$

$$= \int_0^1 1 e^{-x} dx$$

$$= \left[\frac{e^{-x}}{-1} \right]_0^1$$

$$= -e^{-1} + e^0 = 1 - \frac{1}{e}$$

$$= \underline{\underline{0.632}}$$

Q let p be the probability that a coin will fall head in a single toss.

In order to test, H_0 where $H_0: p = \frac{1}{2}$

$$H_1: p = \frac{3}{4}$$

The coin is tossed 5 times and H_0 is rejected if more than 3 heads are obtained.
find the probability of type I error

$$P(X=x) = {}^nC_x p^x (1-p)^{n-x}$$

$$W: x > 3$$

type I error $\Rightarrow$

$$\alpha = P(X \in W | H_0)$$

$$= P(X > 3 | p = \frac{1}{2})$$

$$= P(4) + P(5)$$

$$= {}^5C_4 \left(\frac{1}{2}\right)^4 \left(1 - \frac{1}{2}\right)^1 + {}^5C_5 \left(\frac{1}{2}\right)^5 \left(1 - \frac{1}{2}\right)^0$$

$$= 5 \times \frac{1}{32} + \frac{1}{32}$$

$$= \frac{6}{32}$$

$$\alpha = \frac{3}{16}$$

mean related test of claimants of normality?

t-test

The t-test for the mean is a statistical test for a population mean.

- t-test can be used when the population is normal or nearly normal and sample size $n < 30$.
- The test statistics for the t-test is given by $t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$ with $n-1$ degrees of freedom.

$\bar{x}$ = sample mean

s = sample standard deviation

n = sample size

μ = population mean

Steps (Exam of all steps लिखनी है)

- * First step to write H_0 & H_1
- * then draw graph
- * find critical region & critical value (t_{α})
- * if t in shaded region $\rightarrow$ i.e. null hypothesis false
the null hypothesis is rejected

Steps of t-test

(Solve Step By Step Exam)

- (i) State H_0 and H_1 ,
- (ii) Identify α
- (iii) Identify the degrees of freedom and sketch the graph
- (iv) determine critical values*

- (i) determine critical region
- (ii) find the test statistic, $(t) = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$
- (iii) make the decision

if α not given take $\alpha = 0.05$ Standard Value

for degrees $(n-1)$ n will be given (sample size)

निम्नलिखित प्रकार के (left, Right or two tail test)
 ~~Critical~~ α

* # Steps to determine Critical Value

- (i) determine α & degree of freedom $(n-1)$
- (ii) find the critical values using standard table in the row $n-1$ degrees of freedom

if the hypothesis test is

- (i) left tail, then we consider 1 tail & column with -ve sign
- (ii) Right tail, then we consider 1 tail & column with +ve sign
- (iii) two tail, then we consider 2 tail & column with $\alpha/2$ & a -ve sign

Example Find the critical value for a right test
case for

$$\alpha = 0.01$$

$$\text{and } n = 24$$

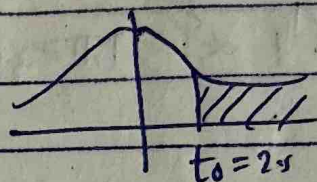
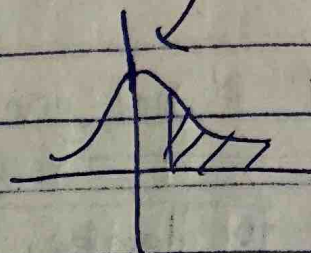
$$\alpha = 0.01$$

$$\text{degree of freedom} = n - 1 = 23$$

By table

$$(t_0 = 2.5)$$

re



Eg. find critical value for two tail test

$$\alpha = 0.10$$

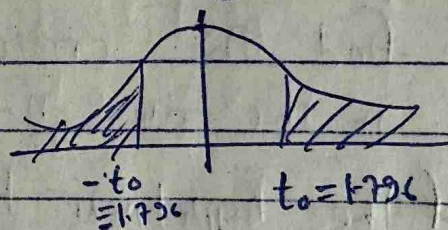
$$n = 12$$

$$\alpha = 0.10$$
$$\text{d.o.f} = n - 1 = 11$$

By table

$$t_0 = 1.796$$

$$-t_0 = -1.796$$



A local telephone company claims that the
average length of a phone call is 8 min.
In a random sample of 18 phone calls,
the sample mean was 7.8 minutes &
the standard deviation was 0.8 min.
Is there enough evidence to support this claim
at $\alpha = 0.05$??

Sol

Hypothesis for mean

ie can use t-test $\Rightarrow$

given
 $\mu = 8$
 $\bar{x} = 7.8$
 $n = 18$
 $s = 0.5$

(i) ~~statistic null & alternative hypothesis~~
 $H_0: \mu = 8$ (given claim)

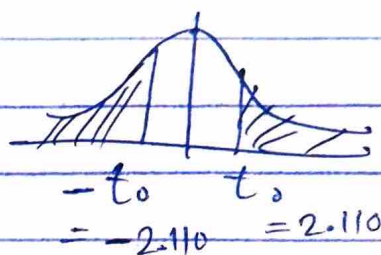
$$H_1: \mu \neq 8$$

(ii) $\alpha = 0.05$

(iii) $\text{dof} = n - 1 = 18 - 1 = 17$

(iv) Critical value $\Rightarrow$

By table $t_0 = 2.110$
 $t_0 = -2.110$] critical values
for $\alpha = 0.05$
in two tailed
& $n = 17$



(v) Determined critical region

(vi) test statistic $= t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{7.8 - 8}{\frac{0.5}{\sqrt{18}}} = -1.70$

Non critical Region,
ie we fail to reject
Null hypothesis,
But we don't have enough
evidence to prove the claim true, ~~ie not enough~~

Since t lies in non critical region we don't
have enough ~~reason~~ to ~~support~~ reject claim

~~we have enough reason to support the claim~~

20/5

Q. A manufacturer claim that its rechargeable batteries have avg 71000 hours

a Random sample of 10 batteries has a mean life of 1002 charges & a standard deviation of 14 charges. Is there enough evidence to support this claim at $\alpha = 0.01$

Sol

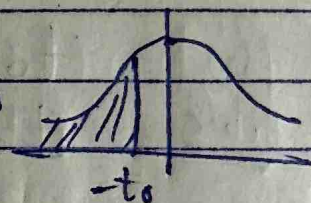
① $H_0: \mu > 1000$

($n=10$)

② $H_1: \mu \leq 1000$

③ $\alpha = 0.01$

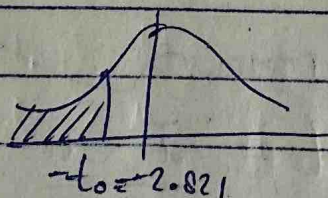
④ $df = n-1 = 10-1 = 9 \rightarrow$ left tail



⑤ critical value

By table, $-t_0 = -2.821$

⑥ Critical Region



⑦ test statistic $t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{1002 - 1000}{14/\sqrt{10}}$

$|t| = 0.45$

⑧

t lies in non-critical Region

$\therefore H_0$ can be accepted

$\therefore$ Claim can be supported or accepted

★ t-test for difference of Means

(25/Nov/24)

Consider two independent samples x_i ($i=1, 2, \dots, n_1$) of size n_1 , and y_j ($j=1, 2, \dots, n_2$) of size n_2 ,

We want to test if the two independent samples have been drawn from two ~~normal~~ normal populations with mean μ_x and μ_y , then the test statistic is $\Rightarrow$

$$t = \frac{(\bar{x} - \bar{y}) - (\mu_x - \mu_y)}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Where $\bar{x}$ & $\bar{y}$ are sample mean $\Rightarrow$

$$\bar{x} = \frac{1}{n_1} \sum x_i$$

$$\bar{y} = \frac{1}{n_2} \sum y_j$$

$$S^2 = \frac{1}{n_1 + n_2 - 2} \left[\sum_i (x_i - \bar{x})^2 + \sum_j (y_j - \bar{y})^2 \right]$$

$$(21) S^2 = \frac{1}{n_1 + n_2 - 2} (n_1 S_1^2 + n_2 S_2^2)$$

Where $S_1^2 = \frac{1}{n_1 - 1} \sum_i (x_i - \bar{x})^2$

$$S_2^2 = \frac{1}{n_2 - 1} \sum_j (y_j - \bar{y})^2$$

follows the t -distribution with $(n_1 + n_2 - 2)$ df

Another
same as
t-test,
~~test~~
But for this
one,
we use
this
t

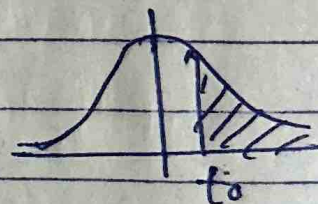
Q Sample of two types of electric Bulbs were tested for length of light, & the following data was obtained.

	I	II	given $n_1 = 8$ $n_2 = 7$ $\bar{x}_1 = 1234$ $S_1 = 36$ $\bar{x}_2 = 1036$ $S_2 = 40$	
Sample size	8	7		
Sample mean	1234	1036		
Sample s.d.	36	40		

Is the difference in the means significant to say the type I is superior to type II regarding the length of light.

Sol as to get Type I (μ) > Type II (μ)
 So taking difference is not significant enough
 (opposite taking to the Que),

① $H_0: \mu_1 = \mu_2$
 $H_1: \mu_1 > \mu_2$ (Right tailed)

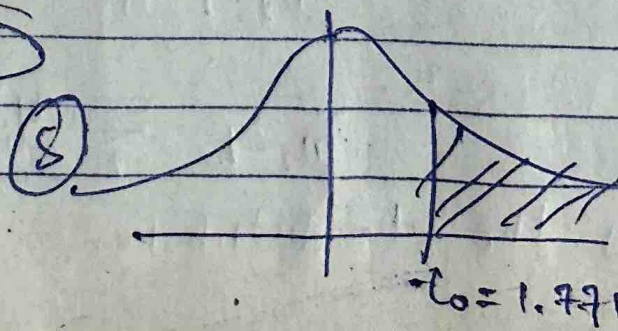


② α (not given) i.e. take standard value
 $\alpha = 0.05$

③ $\text{dof} = n_1 + n_2 - 2$
 $= 7 + 8 - 2$
 $= 13$

④ from table, for $\alpha = 0.05$ & $\text{dof} = 13$

$t_0 = 1.771$



6) test statistics $\Rightarrow$

$$t = \frac{(\bar{x} - \bar{y}) - (\mu_x - \mu_y)}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$\text{where } S = \sqrt{\frac{1}{(n_1 + n_2 - 2)} (n_1 S_1^2 + n_2 S_2^2)}$$

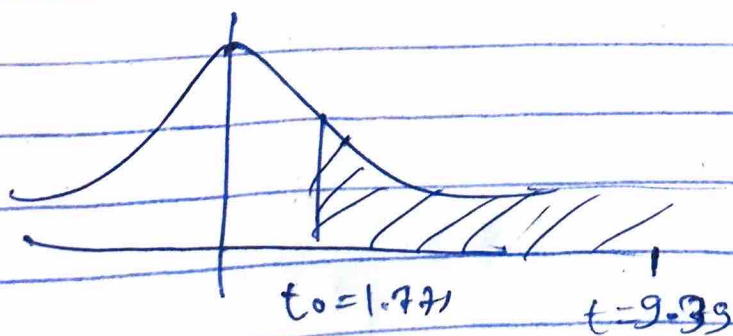
$$S = \sqrt{\frac{1}{(8+7-2)} (8(36)^2 + 7(40)^2)}$$

$$S = 40.7317$$

$$t = \frac{(\bar{x} - \bar{y}) - (\mu_x - \mu_y)}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad \left(\begin{array}{l} \text{took } \mu_0 = \mu_x = \mu_y \\ \text{ie } (\mu_x - \mu_y) = 0 \end{array} \right)$$

$$t = \frac{(1234 - 1036) - (0)}{(40.7317) \sqrt{\frac{1}{8} + \frac{1}{7}}}$$

$$t = 9.39$$



Since t in Critical Region, ie H_0 is Rejected,

ie H_1 is true

ie $(\mu_1 > \mu_2) \in \left(\begin{array}{l} \text{ie type 1 is} \\ \text{superior to} \\ \text{type 2} \end{array} \right)$

Q Of the two salesmen, X claims that he has made more sales than Y.

for the account examined, the following Results

	X	Y
no of sales	10	14
avg size of sales	6200	5600
SD of sales	650	600

Do these Avg size of sales differ significantly?

Sol ① $H_0: \mu_1 = \mu_2$
 $H_1: \mu_1 \neq \mu_2$

② $\alpha = 0.05$ (standard value)

P

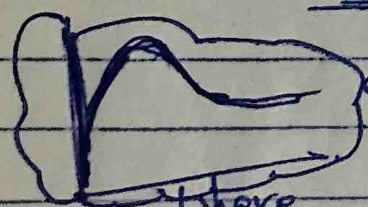
☆ Paired t-test for difference of mean

Refer the ppt

☆ F-test (for comparison of variance of two tests)

The two sample f-test is used to compare two population variances ~~σ_1^2~~ σ_1^2 and σ_2^2 when a sample is randomly selected for each population. The population must be independent & normally distributed.

The test statistic is



$$F = \frac{S_1^2}{S_2^2}$$

$\geq$ (always)

as both variances are

where S_1^2 and S_2^2 represent the sample variances with ~~S_1^2 and~~

$$S_1^2 \geq S_2^2$$

✗ The dof for numerator is $(n_1 - 1) = df_N$
Ans dof for denominator is $(n_2 - 1) = df_D$

where n_1 is size of sample with variance S_1^2
& n_2 is size of sample with variance S_2^2

Steps for using f-test

- ① State H_0 & H_1
- ② identify α
- ③ $df_N = n_1 - 1$ & $df_D = n_2 - 1$
- ④ determine critical value & critical region
- ⑤ find F (test statistics)
- ⑥ Make the decision

for critical value

If $H_1 \Rightarrow$ one tailed, look table corresponding $\alpha = \alpha$
if $H_1 \Rightarrow$ two tailed, then look table for $\alpha = \frac{\alpha}{2}$

Row $n_1 - 1$
Column $n_2 - 1$

① find Critical value for Right tailed
case with $\alpha = 0.05$, with $df_N = 5$
& $df_D = 28$

df_N $\rightarrow$
 df_D $\downarrow$

for $df_N = 5$
 $df_D = 28$
By table for $\alpha = 0.05$

$F_0 = 2.56$ wh

check
 $\alpha = \alpha$

ie $\alpha = 0.05$ in table

one tailed

Q

two tailed

$$\alpha = 0.10$$

$$df_N = 4$$

$$df_D = 6$$

By table

$$\frac{\alpha}{2} = \frac{0.10}{2} = 0.05$$

Column

Row

Q a travel company is marketing brochure ~~indicates~~ indicates the SD of of hotel rate for 2 cities are same. A Random sample of 13 ^{room rate} ~~name~~ ~~is~~ ~~at~~ ~~15~~, ~~29.75~~ in one city has SD of of 27.50 and a random sample of 16 hotel room rates in the other city has a SD of 29.75. Can you reject the agency claim at $\alpha = 0.01$

Sol

$$n_1 = 16, S_1 = 29.75$$

$$n_2 = 13, S_2 = 27.5$$

$$H_0: S_1^2 = S_2^2$$

$$H_1: S_1^2 \neq S_2^2 \text{ (two tailed, use } \frac{\alpha}{2} \text{ table)}$$

$$\alpha = 0.01$$

$$df_N = 15, df_D = 12$$

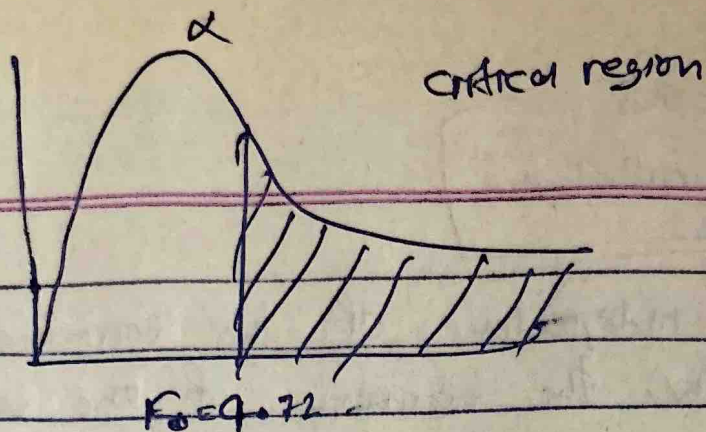
$$\text{critical values for } df_N = 15, df_D = 12 \text{ and } \alpha = 0.005$$

$$F_0 = 4.72$$

from table

$$\alpha = 0.005$$

⑤



⑥
$$F = \frac{S_1^2}{S_2^2} = \frac{(29.75)^2}{(27.5)^2} = 1.17$$

⑦ Since $F = 1.17$ lies in critical Region,
 ie H_0 is Rejected,
 ie H_1 is Accepted

hence

① of Both city are same.

ie we can't Reject claim

⑧ 2 Random Sample drawn from the normal populations are

I 29 16 26 27 23 29 18 29 25 19

II 27 33 42 35 32 34 38 28 41 43 30 37

Test using variance ratio at 5% & 1% level of significance
 whether the populations have the same variance

Chi-square test for independence of attributes

the test

is used to test the independence of two variables, it determines whether the occurrence of one var

2. 3.

2 diff. populations

$H_0 \rightarrow$ fixed Both ~~top~~ independent
 $H_1 \rightarrow$ fixed Both dependent

always
Regardless
of One

always
Right tailed
test

given freq $\Rightarrow$ Observed

$$\chi^2 = \frac{(32 - 30.28)^2}{30.28} + \frac{(13 - 30.28)^2}{30.28} + \frac{(51 - 99.12)^2}{99.12}$$

ANOVA

for multiple
populations

$SSE = ERRO$

$c =$ num of populations

Fratio $\rightarrow \frac{MSE}{SSE/c}$

if $F_c = \frac{MSC}{MSE}$ is MSC > MSE

use F table

